Jordan Brook | CV

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Radar scientist with a particular interest in severe weather and data assimilation. Strong computational and physical sciences background with catastrophe modelling experience for insurance.

Education and Employment

Ongoing Positions

- **Research Scientist, Bureau of Meteorology**
  Radar data assimilation — *2023 – Current*
- **Adjunct Research Fellow, University of Queensland**
  Student supervision and field work — *2023 – Current*

Academic Qualifications

- **Doctor of Philosophy, University of Queensland**
  Atmospheric Sciences, — *2019 – 2023*
- **Bachelor of Science (Honours), University of Queensland**
  Geographical Sciences Major, GPA: 7.0, Thesis awarded a mark of 98% — *2018*
- **Bachelor of Science, University of Queensland**
  Physics and Computational Science Dual Major, GPA: 6.125 — *2015 – 2017*

Academic Awards

- **CARS Research Award**
  Award for enncouraging innovative research directly relevant to the insurance industry, — *2025*
- **Uwe Radok Award**
  Annual award for best PhD thesis in AMOS fields in Australia, — *2024*
- **Dean's Award for Outstanding PhD Thesis**
  Awarded by the University of Queensland and nominated by thesis examiners, — *2024*
- **University of Queensland Excellence Scholarship**
  Awarded based on academic and leadership achievements, — *2015 – 2018*
- **Dean's Commendation for Academic Excellence**
  Recognising outstanding academic performance, — *2015 & 2018*
- **Australian Students Prize**
  Awarded to the 500 highest achieving secondary school graduates, ATAR 99.85, — *2015*

Publications

Journal Articles

- Hu, Y., D. Liu, A. Protat, V. Louf, **J. Brook** and C. Xu, 2026: *Enhancing Production of Synthetic Radar Images from Geostationary Satellite Observations through Generative Diffusion Models*, Artificial Intelligence for the Earth Systems, https://doi.org/10.1175/AIES-D-25-0016.1
- Rennie, S., C. Franklin, S. Cooper, B. Roux, P. Steinle, M. Zidikheri, N. Eizenberg, I. Dharssi, M. Krysta, M. Khanarmuei, F. Smith, A. Smith, N. H. Samrat, C. Griffin, **J. Brook** and C. Bridge, 2026: *ACCESS-A: developing convective-scale numerical weather prediction (NWP) for the Australian continent*, J. South. Hemisphere Earth Syst. Sci., https://doi.org/10.1071/ES25038

- Soderholm, J., M. Kumjian, **J. Brook**, A. Peterson, A. Protat, J. Brimelow, S. Tromel and M. Kunz, 2024: *Measuring hail-like trajectories and growth with the Hailsonde*, Bull. Am. Meterol. Soc., https://doi.org/10.1175/BAMS-D-24-0021.1
- Protat, A., V. Louf and **J. Brook**, 2024: *SWIRL: The First Australian Operational Radar-Based 3D Wind Analysis System*, J. Atmos. Ocean Technol., https://doi.org/10.1175/JTECH-D-23-0155.1
- **Brook, J.**, A. Protat, J. Soderholm, H. McGowan and R. Warren, 2024: *A radar-based hail climatology of Australia*, Mon. Weather Rev., https://doi.org/10.1175/MWR-D-23-0130.1
- A. Guyot, **J. Brook**, A. Protat, K. Turner, J. Soderholm, N. McCarthy and H. McGowan, 2023: *Segmentation of polarimetric radar imagery using statistical texture*, Atmos. Meas. Tech., https://amt.copernicus.org/articles/16/4571/2023/
- **Brook, J.**, A. Protat, J. Soderholm and H. McGowan, 2023: *The Effects of Spatial Interpolation on Dual-Doppler 3D Wind Retrievals*, J. Atmos. Ocean Technol., https://doi.org/10.1175/JTECH-D-23-0004.1
- **Brook, J.**, A. Protat, J. Soderholm, R. Warren and H. McGowan, 2022: *A Variational Interpolation Method for Gridding Weather Radar Data*, J. Atmos. Ocean Technol., https://doi.org/10.1175/JTECH-D-22-0015.1.
- Protat, A., V. Louf, J. Soderholm, **J. Brook** and W. Posonby, 2022: *Three-way calibration checks using ground-based, ship-based, and spaceborne radars*, J. Atmos. Meas. Tech., https://amt.copernicus.org/articles/15/915/2022/amt-15-915-2022.html
- **Brook, J.**, A. Protat, J. Soderholm, J. Carlin, H. McGowan and R. Warren, 2021: *HailTrack - Improving Radar-Based Hailfall Estimates by Modelling Hail Trajectories*, J. Appl. Meteor. Clim., https://doi.org/10.1175/JAMC-D-20-0087.1
- Dowdy, A. , J. Soderholm, **J. Brook**, A. Brown, and H. McGowan, 2020: *Quantifying hail and lightning risk factors based on long-term observations around Australia*, J. Geophys. Res. Atmos., 125, 2020JD033101, https://doi.org/10.1029/2020JD033101
- Soderholm, J.S. , K.I. Turner, **J. Brook**, T. Wedd, and J. Callaghan, 2019: *High-Impact Thunderstorms of the Brisbane Metropolitan Area*, J. South. Hemisphere Earth Syst. Sci., 69, 239-251, https://www.publish.csiro.au/es/es19017
- Downes, S.M., C. Langlais, **J. Brook**, and P. Spence, 2017: *Regional Impacts of the Westerly Winds on Southern Ocean Mode and Intermediate Water Subduction*, J. Phys. Oceanogr., 47, 2521–2530, https://doi.org/10.1175/JPO-D-17-0106.1

Conference Presentations

- **Brook, J.**, 2026: *Panel Discussion: Best Practice Vs. Best Guess,*, Invited Speaker, Catastrophe and Reinsurance Symposium 20 Mar., Manly, Australia.
- **Brook, J.**, 2026: *Uwe Radok Award Plenary Presentation,*, AMOS Annual Conference 16-20 Feb., Hobart, Australia.
- **Brook, J.**, C. Bishop, A. Protat, 2025: *Pre-Emptive Forecasting with the Ensemble Transform Kalman Filter,*, International Symposium on Data Assimilation 29 Sep-3 Oct., Melbourne, Australia.
- **Brook, J.**, C. Bishop, A. Protat, 2025: *Pre-Emptive Forecasting: A new approach for rapidly updating weather forecasts*, 2nd science-industry workshop on severe convective storms, 7-8 Aug., Sydney, Australia.
- **Brook, J.**, C. Bishop*, A. Protat, 2024: *Pre-Emptive Forecasting with the Ensemble Transform Kalman Filter*, International Symposium on Data Assimilation 21-25 Oct., Kobe, Japan. **Presenting Author*.
- **Brook, J.**, A. Protat, J. Soderholm, C. Potvin, H. McGowan, 2024: *Beyond the blur: Variational techniques for radar gridding and 3D wind retrieval*, AMS Radar Conference 28 Aug.-01 Sep., Minneapolis, USA
- **Brook, J.**, A. Protat, J. Soderholm, H. McGowan, 2023: *Radar perspectives on thunderstorm dynamics and associated hazards*, Science-industry workshop on severe convective storms, 3-4 Aug., Sydney
- **Brook, J.**, A. Protat, M. Curtis, J. Soderholm, 2024: *Recent advances in radar-based 3D wind retrievals*, AMOS Annual Conference 5-9 Feb., Canberra
- **Brook, J.**, A. Protat, J. Soderholm, H. McGowan, R. Fortune, 2022: *A radar-based Hail Climatology of Australia*, AMOS Annual Conference 28 Nov. - 1 Dec., Adelaide
- **Brook, J.**, A. Protat, J. Soderholm, H. McGowan, R. Fortune, 2022: *A radar-based Hail Climatology of*

Australia, North American Workshop On Hail & Hailstorms 20 - 22 September, Online

- **Brook, J.**, A. Protat, J. Soderholm, V. Louf, H. McGowan, 2022: *Unlocking Unprecedented Detail in Weather Radar Imagery*, 11th European Radar Conference, 29 Aug. - 2 Setp., Locarno, Switzerland.
- **Brook, J.**, A. Protat, J. Soderholm, H. McGowan, R. Warren, 2022: *A Variational Interpolation Method for Gridding Weather Radar Data*, 13th International Conference on Southern Hemisphere Meteorology and Oceanography 8 - 12 February, Online
- **Brook, J.**, A. Protat, J. Soderholm, V. Louf, H. McGowan, 2021: *Unlocking Unprecedented Detail in Weather Radar Imagery*, Annual BoM Research and Development Workshop, 8 - 11 November, Online. *Winner: Best Poster Presentation*
- **Brook, J.**, A. Protat, J. Soderholm, H. McGowan, R. Warren, 2019: *HailTrack: Modelling Hail Trajectories Using Weather Radar Data*, 39th AMS Radar Conference, 16 - 20 September, Nara, Japan. *Runner up for Best Student Poster*
- **Brook, J.**, A. Protat, J. Soderholm, J. Carlin, H. McGowan, 2018: *Computationally Estimating Hail Trajectories*, North American Workshop On Hail & Hailstorms, 14 - 16 August, Boulder, CO., USA

Relevant Experience

Field Campaigns

- **Research Voyage IN2019V06**
  Research Vessel Investigator *Oct 2019 – Dec 2019*

  I was a part of a six-week voyage aimed at taking observations of the daily cycles of convective storms in Australia's far northern waters. My role was to take periodic atmospheric soundings and assist with radar operations aboard the ship.

- **Large Hail Formation and Trajectories Campaign (LIFT)**
  German Research Foundation *Jun 2024 – Jul 2024*

  I was an invited participant of this field campaign aimed at collecting in-situ observations of hail trajectories using world-first "Hailsonde" instruments.

- **Bistatic radar pilot project in southeast Queensland**
  University of Queensland *Jan 2023 – Jan 2024*

  Required to organize and implement the installation of bistatic radar equipment at the Moreton Bay Research Station and UQ St Lucia campus.

- **In-situ Collaborative Exp. for the Collection of Hail In the Plains (ICECHIP)**
  National Science Foundation *June 2025 – July 2025*

  Invited participant of the HailSonde team required to release the second generation of HailSonde devices into hailstorms in the United States.

Academic References

- **Prof. Hamish McGowan**
  Professor, School of Earth and Environmental Sciences, UQ *h.mcgowan@uq.edu.au*
- **Dr. Alain Protat**
  "Radar Science and Nowcasting" Team Leader, The Bureau of Meteorology *alain.protat@bom.gov.au*